From Style to Prompt: Creating Images with an Art Historian's Eye

Teaching with AI (TAI)

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Subject: Art styles and image prompting
Activity duration: 90 min

Steps:

1. Art Styles (30 minutes)
    - Motivation + Assignment 1: spot the difference (10 min)
    - Assignment 2: mystery image (15 min)
    - Assignment 3: Real or fake (5 min)
2. Prompting images (60 minutes)
    - Assignment 4: reverse engineer your mystery image (15 min)
    - Assignment 5: swap and guess (10 min)
    - Assignment 6: create a logo (30 min)

Learning goals:

- Art styles have common visual features.
- Distinguish art styles in paintings and photographs.
- Using AI to learn responsibly: describe images, request verification links from museums, and cross-check facts
- How to prompt to obtain the image you want
- Create a logo

Setup:

- Students need access to an AI LLM chat bot, and an AI image generation tool.
- The teacher provides and prints mystery artworks for every pair or students.
- The teacher prints a Style Feature Guide and Style Card for every pair of students.

From style to prompt

Creating images with an art historian's eye

Modern AI image generators can create images in all kind of styles. For example, since they have been trained on many famous artworks, they can therefore reproduce the style of famous artists or art styles. This gives anyone the chance to, without being a master painter, create an image of their wish in the style of, let's say, Vincent van Gogh. We give the style as part of a textual prompt to the model, and it creates an image.
This creates an opportunity, but is also a challenge: we need to be able to *describe the image we want to create in words*. Sometimes, it is not easy to express in words why we like a painting, photograph or drawing. In this lesson, you will learn about different art styles and their main visual features. Then, you will practice with creating your own images with AI.

Assignment 1: Spot the difference

Your teacher will show two paintings and two photographs. Also, you will receive a Style Guide. Work with a partner to identify what makes the pairs of paintings and photographs different in style. Use the Style Guide to make the differences concrete.

Pair A: paintings

Question 1: Come up with a name for the paintings.

- Name for Painting 1

- Name for Painting 2

Question 2: Describe the differences between the paintings in your own words.

Question 3: Look at the Style Guide and find two style features present in Painting 1 but not in Painting 2.

Question 4: Now find two style features present in Painting 2 but not Painting 1.

Pair B: photographs

Question 5: Come up with a name for the photos.

- Name for Photo 1
- Name for Photo 2

Question 6: Describe the differences between the photographs in your own words.

Question 7: Look at the Style Guide and find two style features present in Photo 1 but not in Photo 2.

Question 8: Now find two style features present in Photo 2 but not Photo 1.

1 Art Styles

Some combinations of style features are often used together. This gives collections of paintings that are often made in a certain period of time a similar look. These looks are described by art styles like realism, impressionism or Art Nouveau, for example.
In the next assignment, you will try to figure out the art style of a given piece of art, by first identifying the style features.

Assignment 2: Mystery art work

a) As a group of 2-3 students you receive an image of a mystery art work and an empty style card. Fill out the style card for your assigned mystery art work. Afterwards, answer the questions below.

Question 9: How did you use AI to fill out your style card?

Question 10: How did you verify that the information that the AI gave you was correct?

b) Write the name of the art style on your given mystery image and collect all of them in the front of the class. Take a look at the different images and styles.

Question 11: Which painting and style do you like, and why? Describe it in terms of aspects in the Style Guide.

Assignment 3: Real or fake?

Imagine that someone finds an old painting in their attic, inherited from an old relative and forgotten. Is it the long lost Rembrandt that no one ever found? Or, is it a fake? For questions like this, art connoisseurs play an important role. They know everything about art styles and art history, and are often even specialized in a specific style or age. There are several techniques to determine the origin of an old painting: comparing it to other paintings, doing chemical analysis of the paint, trying to figure out the line of prior possessors of the art work.

Question 12: How could AI help in the question whether this art piece is real or fake?

Question 13: Which aspects of this research will have to be done by humans?

Question 14: What are limitations of AI as it comes to evaluating art?

2 Image generation with AI

The rest of this lesson we will generate images ourselves. We will create a personalized prompting template that we can use in the future, to help us create the images we want.

Assignment 4: Reverse engineer your mystery artwork

a) Open an AI Image generation program. Without using the name of the artist and artwork, try to recreate the piece of art that you received as a mystery artwork. Make use of the style features you collected in the previous exercise, and try to describe what you see as well as possible.

Question 15: Paste your prompt here.

b) You will now receive a Prompt Guide from your teacher. Repeat the exercise and rewrite/improve your prompt using the tips.

Question 16: Paste your new prompt here.

Question 17: Compare the images created by the two versions of the prompt. What are the most important differences?

Question 18: What tip did you find useful?

Question 19: What would you add to the tips? Add them to your personalized Prompt Guide.

c) Finally, put the best version of your prompt in an AI chat bot, and ask it to help you improve it even more. Say that you want to create the mystery painting (title, artist) without using the title and artist.

Question 20: Paste the final version of your prompt here.

Question 21: Ask your AI Chatbot for more tips to write prompts for images and add the ones you like to your personal Prompt Guide.

Assignment 5: Assignment 6: Swap and guess

a) Describe a scene from a book, movie or TV series that you like. You will use this later to generate an image.

b) Generate an image of the scene you described, using your personal Prompt Guide. Each student generates one. Save your prompt and the image.

Question 22: Write your prompt here.

c) Exchange images with a neighbor. Try to guess the prompt that your neighbor used.

Question 23: Write your guess prompt for your neighbors image here.

d) Try out the guess prompt and compare it to the original image. Also compare the prompts. Discuss what details were essential or missing.

Question 24: Which parts of the prompt did you miss?

Question 25: Why is it impossible to recreate the image exactly?

Assignment 6: Logo creation

a) In teams of 2 or 3, create a logo for the Teaching in AI (TAI) project. We will vote for the best logo, which will be used on the website and the publication of all the teaching material of this project. The logo will be seen by all the teachers and students learning from this material like you do now.

Tips:

- Consider the target audience and think about what they would like.
- You can chat with an LLM to come up with ideas, that you can then translate into a prompt.

Afterwards, answer the following question.

Question 26: Which aspects of today's lesson did you use in the design of your logo?